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High Impact Polystyrene (HIPS)

HIPS is a grade of PS. The graph bars on the material properties cards below compare HIPS to: styrenic plastics (top), all thermoplastics (middle), and the entire database (bottom). A full bar means this is the highest value in the relevant set. A half-full bar means it's 50% of the highest, and so on.

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Mechanical Properties

Elastic (Young's, Tensile) Modulus

Elongation at Break

1.9 GPa

0.28×10^6 psi

40 %

Flexural Strength

Impact Strength: Notched Izod

62 MPa

9.0×10^3 psi

45 J/m

0.85 ft-lb/in

Poisson's Ratio

Tensile Strength: Ultimate (UTS)

0.41

32 MPa

4.6×10^3 psi

Thermal Properties

Glass Transition Temperature

Maximum Temperature: Autoignition

100 °C

210 °F

420 °C

780 °F

Maximum Temperature: Decomposition

Maximum Temperature: Mechanical

330 °C

620 °F

70 °C

160 °F

Specific Heat Capacity

Thermal Conductivity

1400 J/kg-K

0.33 BTU/lb-°F

0.22 W/m-K

0.13 BTU/h-ft-°F

Thermal Expansion

Vicat Softening Temperature

80 $\mu\text{m/m-K}$

110 °C

220 °F

Electrical Properties

Dielectric Strength (Breakdown Potential)

Electrical Resistivity Order of Magnitude

18 kV/mm

0.7 V/mil

14 $10^x \Omega\text{-m}$

Otherwise Unclassified Properties

Density

Limiting Oxygen Index (LOI)

1.0 g/cm³

65 lb/ft³

18 %

Water Absorption After 24 Hours

0.080 %

Common Calculations

Stiffness to Weight: Axial

Stiffness to Weight: Bending

1.0 points

40 points

Strength to Weight: Axial

Strength to Weight: Bending

8.5 points

Thermal Diffusivity

0.15 mm²/s

22 points

Thermal Shock Resistance

15 points

Followup Questions

How are the material properties defined?

How do I compare this exact variant to another material?

Further Reading

ASTM D883: Standard Terminology Relating to Plastics

SPI Plastics Engineering Handbook of the Society of the Plastics Industry, Inc., 5th ed., Michael L. Berins (editor), 2000

Modern Plastics Handbook, Charles A. Harper (editor), 1999

Plastics Materials, 7th ed., J. A. Brydson, 1999

